

Review of Last Lecture

6.1 Introduction

6.2 Discrete-Time Signals – Sequences

6.3 Mathematical Model of Discrete-Time Systems

**6.4 Classical Time-Domain Solution of Linear
Constant-Coefficient Difference Equations**

Continuous-Time System

- Differential Equation
 - Convolution Integral
 - Laplace Transform
 - Fourier Transform
-
- System Function
 - Convolution Theorem

vs.

Discrete-Time System

- Difference Equation
- Convolution Sum
- z-Transform
- Discrete-Time Fourier Transform, Discrete Fourier Transform
- System Function
- Convolution Theorem

6.2 Discrete-Time Signals - Sequences



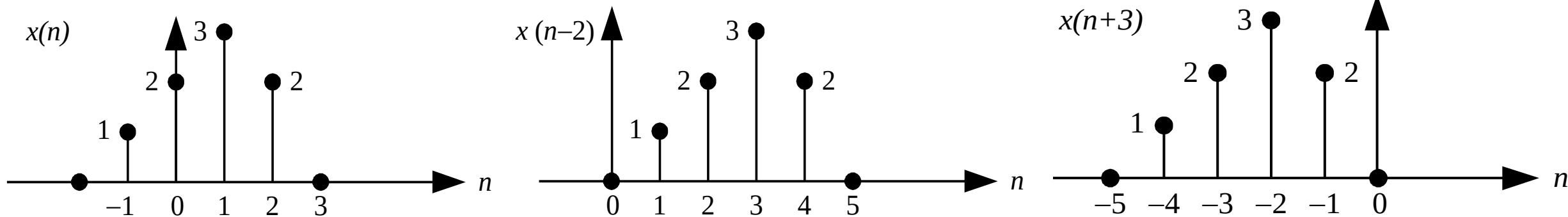
6.2.2 Operations on Discrete-Time Signals

1. Shifting of Sequences

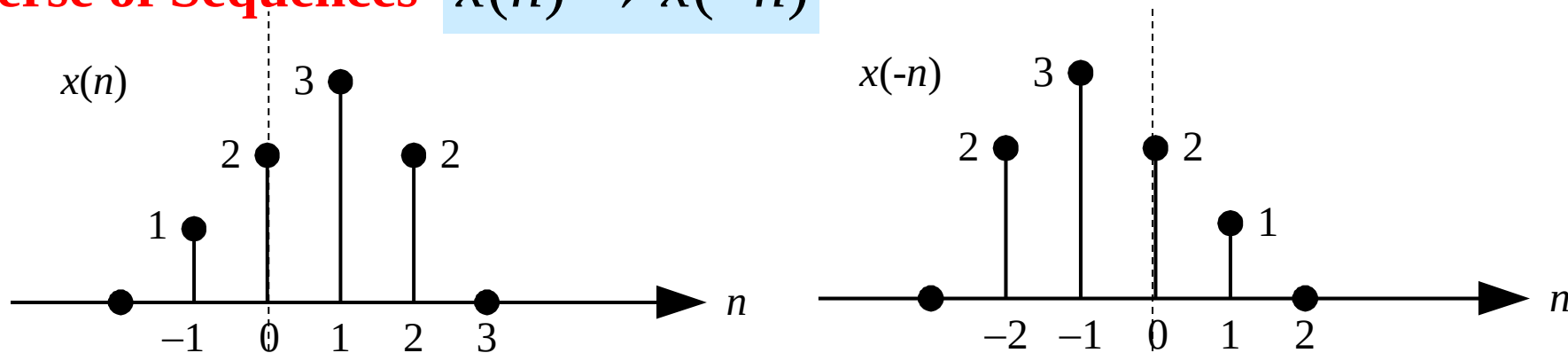
$x(n-m)$ means shifting $x(n)$ to the right by m units.

$x(n+m)$ means shifting $x(n)$ to the left by m units.

$m > 0$



2. Reverse of Sequences $x(n) \rightarrow x(-n)$



Given sequence $x(n) = \{-2, 1, 2, \overset{\downarrow}{0}, 3, 2, 4, 6, 5\}$, then $x(2n) = (\quad)$

A $\{-2, \overset{\downarrow}{2}, 3, 4, 5\}$

B $\{1, \overset{\downarrow}{0}, 2, 6\}$

C $\{2, \overset{\downarrow}{0}, 3, 2\}$

D $\{-2, 1, \overset{\downarrow}{0}, 2, 6\}$

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6.2 Discrete-Time Signals - Sequences



3. Scaling of Sequences

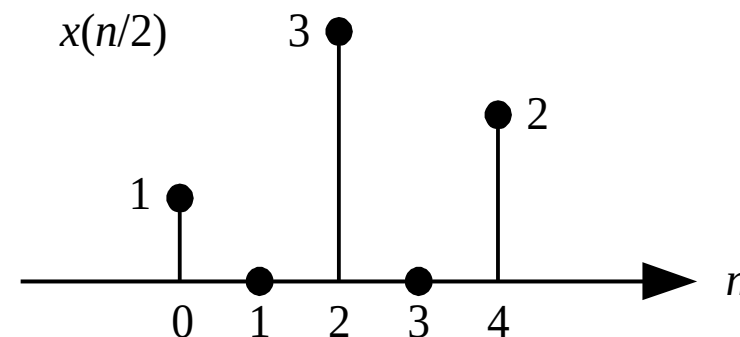
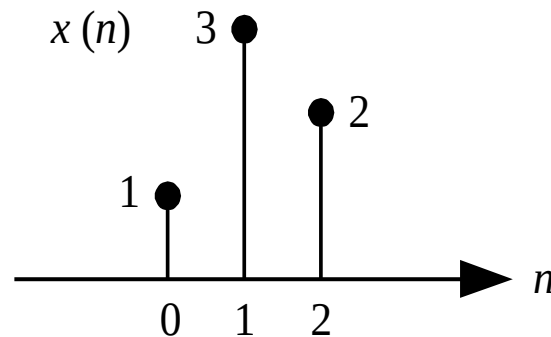
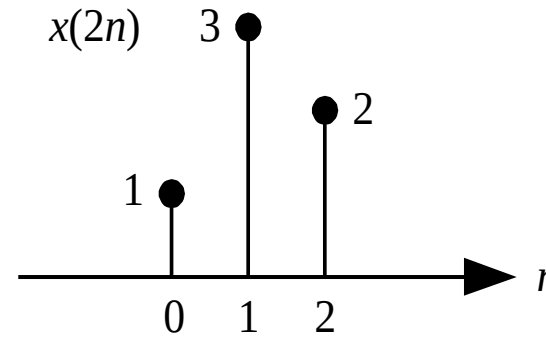
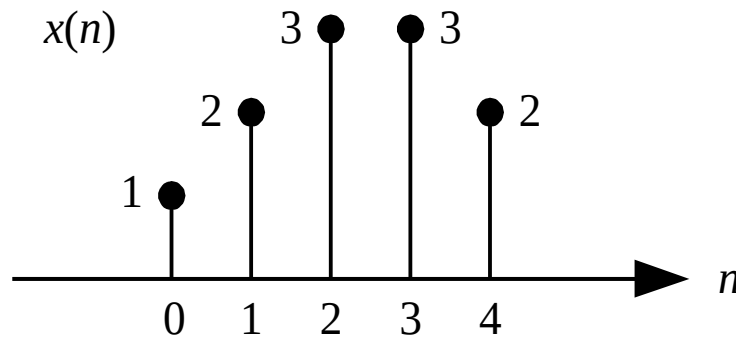
$a > 0$ is a positive integer.

$$x(n) \rightarrow x(an)$$

$$x(n) \rightarrow x\left(\frac{n}{a}\right)$$

Compression: Extract one point every $(a-1)$ points around the origin.

Expansion: Insert $(a-1)$ zero-valued points between adjacent points.



6.3 Mathematical Model of Discrete-Time Systems



Mathematical Model of Continuous Systems—Differential Equation

$$C_0 \frac{d^n r(t)}{dt^n} + C_1 \frac{d^{n-1} r(t)}{dt^{n-1}} + \dots + C_n r(t) = E_0 \frac{d^m e(t)}{dt^m} + E_1 \frac{d^{m-1} e(t)}{dt^{m-1}} + \dots + E_m e(t)$$

Basic Operations: Differentiation, Multiplication by Coefficients, Addition

Mathematical Model of Discrete Systems—Difference Equation

$$y(n) = -\sum_{k=1}^N a_k y(n-k) + \sum_{r=0}^M b_r x(n-r)$$

The input is the discrete sequence $x(n)$ and its time-shifted functions
 $x(n), x(n-1), x(n-2), \dots$

The output is the discrete sequence $y(n)$ and its time-shifted functions

$$y(n), y(n-1), y(n-2), \dots$$

Basic Operations: **Delay**, Multiplication by Coefficients, Addition

6.4 Classical Time-Domain Solution of Linear Difference Equations

The complete solution of a difference equation, i.e., the complete response of the system, consists of the homogeneous solution $y_h(n)$ and the particular solution $y_p(n)$:

$$y(n) = y_h(n) + y_p(n)$$

The form of the homogeneous solution $y_h(n)$ is determined by the characteristic roots of the homogeneous equation.

The form of the particular solution $y_p(n)$ is determined by the simplified form of the right-hand side of the equation after substituting the excitation signal.

1. Find the Homogeneous Solution (Inherent Response of the System)

Homogeneous equation of the difference equation:: $\sum_{k=0}^N a_k y(n-k) = 0$

$$a_0 \alpha^N + a_1 \alpha^{N-1} + \cdots + a_k \alpha^{N-k} + \cdots + a_N = 0$$

Characteristic Equation

The roots of the equation $\alpha_1, \alpha_2, \alpha_3 \dots$

Characteristic Roots

α_i is called the natural frequency (or free frequency) of the system.

6.4 Classical Time-Domain Solution of Linear Difference Equations

Homogeneous Solutions of Difference and Differential Equations

Characteristic Roots	Homogeneous Solution of Difference Equation	Homogeneous Solution of Differential Equation
Distinct Real Roots $\alpha_1, \alpha_2, \dots, \alpha_N$	$\sum_{k=1}^N C_k \alpha_k^n$	$\sum_{k=1}^N A_k e^{\alpha_k t}$
k -fold Root α_1	$\sum_{i=1}^k C_i n^{k-i} \alpha_1^n$	$\sum_{i=1}^k A_i t^{k-i} e^{\alpha_1 t}$
Conjugate Complex Roots $\alpha \pm j\beta$	$\left(\sqrt{\alpha^2 + \beta^2} \right)^n \left[P \cos(n\omega_0) + Q \sin(n\omega_0) \right]$ $\rho = \sqrt{\alpha^2 + \beta^2} \quad P = C_1 + C_2 \quad Q = j(C_1 - C_2)$	$e^{\alpha t} (A \cos \beta t + B \sin \beta t)$

The undetermined coefficients of the homogeneous solution are determined by the **boundary conditions** (after finding the particular solution).

6.4 Classical Time-Domain Solution of Linear Difference Equations

2. Find the Particular Solution: Determined by the Form of the Excitation

Free Term (Right-Hand Side of the Equation)	Form of Particular Solution
α^n	$D\alpha^n$ (if α is not a characteristic root) $Dn^r\alpha^n$ (if α is an r -fold characteristic root)
n^k	$D_0n^k + \dots + D_{k-1}n + D_k$ (if 1 is not a characteristic root) $n^r[D_0n^k + \dots + D_{k-1}n + D_k]$ (if 1 is an r -fold characteristic root)
$\alpha^n n^k$	$[D_0n^k + \dots + D_{k-1}n + D_k]\alpha^n$ (if α is not a characteristic root) $n^r[D_0n^k + \dots + D_{k-1}n + D_k]\alpha^n$ (if α is an r -fold characteristic root)
$\cos(n\omega)$ or $\sin(n\omega)$	$D_1\cos(n\omega) + D_2\sin(n\omega)$ (if $e^{\pm j\omega}$ is not a characteristic root)
$\alpha^n \cos(n\omega)$ or $\alpha^n \sin(n\omega)$	$\alpha^n [D_1\cos(n\omega) + D_2\sin(n\omega)]$ (if $\alpha e^{\pm j\omega}$ is not a characteristic root)

➤ By observing the function form of the **free term** on the right-hand side of the equation, try to **select the form of the particular solution**.

➤ Substitute the particular solution into the equation and use the method of undetermined coefficients to determine D_i .

6.4 Classical Time-Domain Solution of Linear Difference Equations

3. Determine the Undetermined Coefficients of the Homogeneous Solution from the Boundary Conditions

General Form of the Complete Solution (for Distinct Roots):

$$y(n) = C_1\alpha_1^n + C_2\alpha_2^n + \cdots + C_N\alpha_N^n + D(n) = \sum_{k=1}^N C_k\alpha_k^n + D(n)$$

C_i must be determined using N given boundary conditions. For example:

$$\left\{ \begin{array}{l} y(0) = C_1 + C_2 + \cdots + C_N + D(0) \\ y(1) = C_1\alpha_1 + C_2\alpha_2 + \cdots + C_N\alpha_N + D(1) \\ \text{.....} \\ y(N-1) = C_1\alpha_1^{N-1} + C_2\alpha_2^{N-1} + \cdots + C_N\alpha_N^{N-1} + D(N-1) \end{array} \right.$$

For causal systems (where the excitation is applied at $n=0$), $y(-1), y(-2), \dots, y(-N)$ are often given as boundary conditions.

Disadvantages of the Classical Method:

1. If the excitation term on the right side of the difference equation is relatively complex, it is difficult to handle.
2. If the excitation changes, the entire solution process must be repeated.
3. If the initial conditions change, the entire solution process must be repeated.
4. This method is a purely mathematical approach and cannot highlight the physical concepts of system response.

$$y(n) = \sum_{k=1}^N C_k \alpha_k^n + D(n)$$

Contents of This Lecture

6.5 Zero-State Response and Zero-Input

Response of Discrete-Time Systems

6.6 Unit Sample Response of Discrete-Time Systems

6.7 Convolution Sum

Learning Objectives of This Lecture

1. Familiarize yourself with the classical time-domain solution method for difference equations and use **boundary conditions** correctly;
2. Familiarize yourself with the solution of **the unit sample response**;
3. Master the use of **convolution sum** to find the **zero-state response** proficiently.



Complete Solution = Homogeneous Solution + Particular Solution



Total Response = Natural Response + Forced Response

Total Response = Zero-Input Response + Zero-State Response

$$\begin{aligned} y(n) &= \sum_{k=1}^N C_k \alpha_k^n + D(n) \\ &\quad \downarrow \qquad \qquad \downarrow \\ &\quad \text{Natural} \quad \text{Forced} \\ &\quad \text{Response} \quad \text{Response} \\ &= \sum_{k=1}^N C_{zik} \alpha_k^n + \sum_{k=1}^N C_{zsk} \alpha_k^n + D(n) \\ &\quad \downarrow \qquad \qquad \underbrace{\qquad \qquad \qquad} \\ &\quad \text{Zero-Input} \quad \text{Zero-State} \\ &\quad \text{Response} \quad \text{Response} \end{aligned}$$



1. Zero-Input Response

➤ **Definition of Zero-Input Response:** The response generated solely by the initial state of the system (with the input signal being zero), denoted as $y_{zi}(n)$.

➤ Mathematical Model:
$$\sum_{k=0}^N a_k y(n-k) = 0$$

➤ Solution Method: Determine the form of the zero-input response based on the characteristic roots of the difference equation, then solve for the **undetermined coefficients of the homogeneous solution** using **boundary conditions**.



2. Zero-State Response

➤ **Definition of Zero-State Response:** The zero-state response generated by the excitation $x(n)$ when the initial state of the system is zero, denoted as $y_{zs}(n)$.

➤ Mathematical Model:
$$\sum_{k=0}^N a_k y(n-k) = \sum_{r=0}^M b_r x(n-r)$$

➤ Solution :
$$y_{zs}(n) = \sum_{k=1}^N C_{zsk} \alpha_k^n + D(n)$$

For causal systems (the excitation signal is applied at $n=0, y(-1), y(-2), \dots, y(-N)$) are usually given as boundary conditions.

The zero state means $y(-1), y(-2), y(-3), \dots, y(-N)$ are all zero.

$y(0), y(1), y(2), \dots, y(N-1)$ can be obtained by the iterative method.



For the N -th-order difference equation, to determine the coefficients of the homogeneous solution for the **zero-input response** or **zero-state response**, boundary conditions (i.e., N values of $y(n)$) are required.

➤ Determining coefficients of the homogeneous solution for **zero-input response**:
The n in the initial condition $y(n)$ can be either **positive or negative** and is **not necessarily continuous**. $y(-1), y(-2), \dots, y(-N)$ or $y(0), y(1), \dots, y(N-1)$ can all be used as **boundary conditions**. For $y(n)$ when $n \geq 0$, it can be obtained by iteratively substituting $y(n)$ ($n < 0$) into the difference equation, and **the right-hand side of the difference equation must be set to zero during iteration**.

➤ Determining coefficients of the homogeneous solution for **zero-state response**:
Let $y(-1)=y(-2)=\dots=y(-N+1)=0$, and iteratively find $y(0), y(1), \dots$. **Assuming the excitation is applied at $n = 0$, the boundary conditions must include at least one $y(n)$ when $n \geq 0$, and n must be continuous when $n < 0$** (to ensure $y(n)$ with $n \geq 0$ can be obtained by iteration). For example, $y(0), y(-1), \dots, y(-N+1)$ can be used as **boundary conditions**.



Example 6-5: Given the input signal $x(n]=u(n)$ and the difference equation of a discrete-time LTI system:

$$y(n) - 0.9y(n-1) = 0.05x(n)$$

- (1) Find the zero-state response of the system.
- (2) If the boundary condition $y(-1)=1$, find the zero-input response and the total response.

Solution:

- (1) The homogeneous solution of the equation is $C(0.9)^n$.

According to the form of the excitation on the right-hand side of the equation, the particular solution is chosen as D .

Substitute the particular solution into the difference equation: $D - 0.9D = 0.05$, $D = 0.5$

The form of the complete solution is: $y(n) = C(0.9)^n + 0.5$

From $y(-1)=0$, use the **iterative method** to get $y(0) = 0.9y(-1)+0.05 = 0.05$

Substitute $y(0)$ into $y(n)$: $0.05 = C + 0.5 \Rightarrow C = -0.45$

$$\therefore y_{zs}(n) = \left[-0.45 \times (0.9)^n + 0.5 \right] u(n)$$



(2) Find the **zero-input response**:

Set the excitation to zero, and the difference equation is:

$$y(n) - 0.9y(n-1) = 0$$

The zero-input response is $y_{zi}(n) = C_{zi} \times (0.9)^n$

Substitute $y(-1) = 1$: $1 = C_{zi} \times (0.9)^{-1} \Rightarrow C_{zi} = 0.9$

Total response: $y(n) = y_{zs}(n) + y_{zi}(n)$

$$= \underbrace{\left[-0.45 \times (0.9)^n + 0.5 \right]}_{\text{zero-state response}} + \underbrace{0.9 \times (0.9)^n}_{\text{zero-input response}}$$

$$= \underbrace{0.45 \times (0.9)^n}_{\text{free response}} + \underbrace{0.5}_{\text{forced response}} \quad (n \geq 0)$$



Finding the Complete Response by the Classical Method

The homogeneous solution of the equation is $C(0.9)^n$

According to the form of the excitation on the right-hand side of the equation, the particular solution is chosen as **D** .

Substitute the particular solution into the difference equation:

$$D - 0.9D = 0.05, \quad D = 0.5$$

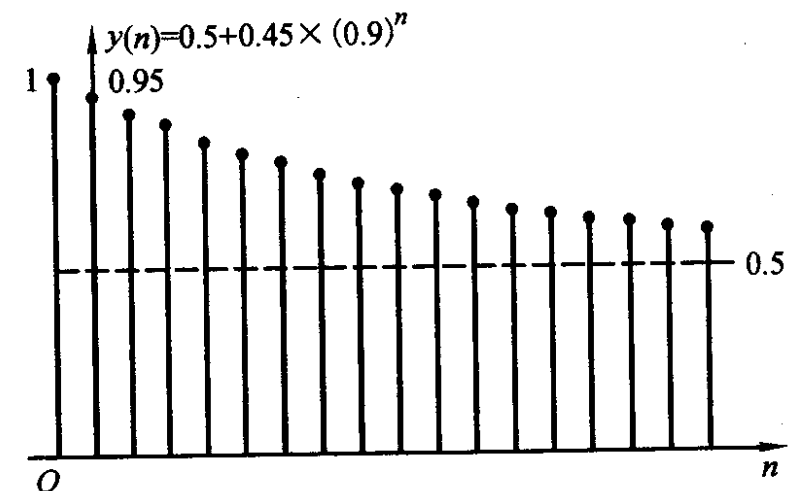
The form of the complete solution is: $y(n) = C(0.9)^n + 0.5$

From $y(-1)=1$, substituting into the equation gives $y(0) = 0.9y(-1)+0.05 = 0.95$

Substitute **$y(0)=0.95$** into the complete solution **$y(n)$** :

$$0.95 = C + 0.5 \Rightarrow C = 0.45$$

$$\therefore y(n) = \left[0.45 \times (0.9)^n + 0.5 \right] u(n)$$



The result is the same as before!

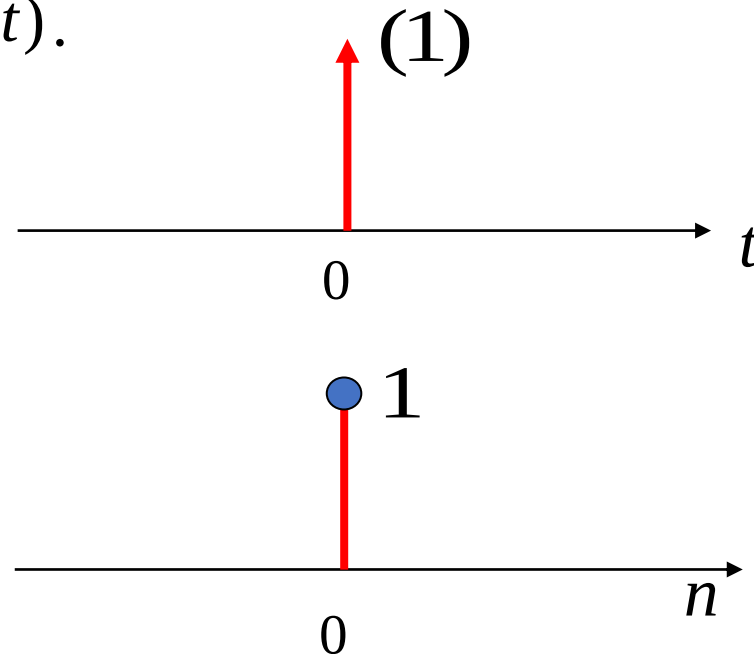
6.6 Unit Sample Response of Discrete-Time Systems

Unit Sample Response: It is the **zero-state response** of the system when the excitation is $\delta(n)$, denoted by $h(n)$.

Note the difference between $\delta(n)$ and $\delta(t)$.

$$\begin{cases} \int_{-\infty}^{\infty} \delta(t) dt = 1 & (t = 0) \\ \delta(t) = 0 & (t \neq 0) \end{cases}$$

$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



Solution of Unit Sample Response: Convert the unit sample excitation signal into boundary conditions.

Convert $\delta(n)$ into boundary conditions, so finding the unit sample response is converted into finding **the homogeneous solution**, i.e., **the zero-input response when $n > 0$** .

6.6 Unit Sample Response of Discrete-Time Systems

Example 6-6: Given the system's difference equation, find the unit sample response.

$$y(n) - 3y(n-1) + 2y(n-2) = x(n)$$

Solution: $h(n)$ satisfies the equation

$$h(n) - 3h(n-1) + 2h(n-2) = \delta(n)$$

(1) Find Zero-Input Response

$$\alpha^2 - 3\alpha + 2 = 0, \alpha_1 = 1, \alpha_2 = 2$$

The homogeneous solution is: $C_1 + C_2 2^n$

(2) Boundary Conditions

For a causal system, $h(-1) = 0, h(-2) = 0$

By substituting into the difference equation, we can derive:

$$h(0) = 3h(-1) - 2h(-2) + \delta(0) = 1$$

$$h(1) = 3h(0) - 2h(-1) + \delta(1) = 3$$

6.6 Unit Sample Response of Discrete-Time Systems

$$C_1 + C_2 2^n$$

Take $h(0) = 1, h(1) = 3$ as the boundary conditions for **the zero-input response**

$$\begin{cases} C_1 + C_2 = 1 \\ C_1 + 2C_2 = 3 \end{cases} \longrightarrow C_1 = -1 \quad C_2 = 2$$

(3) Find Unit Sample Response

$$h(n) = (-1 + 2 \cdot 2^n) u(n) = (2^{n+1} - 1) u(n)$$

Note: The basic principle for choosing boundary conditions is that **the effect of $\delta(n)$ must be reflected in the boundary conditions.**

6.6 Unit Sample Response of Discrete-Time Systems

Analyzing Causality and Stability of Systems Based on Unit Sample Response

Causal System: A system where the output change does not precede the input change. That is, the response depends only on the current and previous excitations.

The **necessary and sufficient condition** for a discrete-time linear time-invariant system to be **a causal system** is:

$$h(n) = 0 \ (n < 0) \text{ or } h(n) = h(n)u(n)$$

Stable System: If the input is bounded, the output must be bounded.

The **necessary and sufficient condition** for a discrete practical linear time-invariant system to be **a stable system** is:

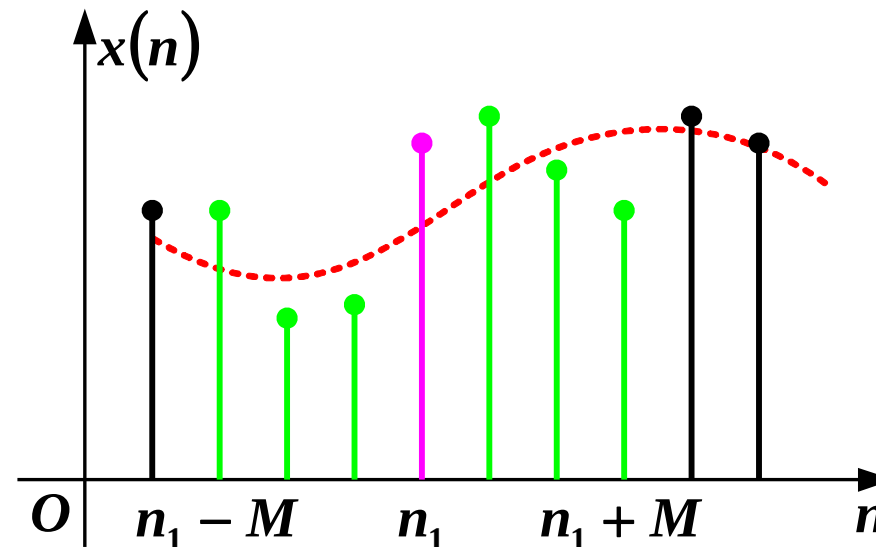
$$\sum_{n=-\infty}^{\infty} |h(n)| \leq M, M \text{ is a bounded positive value}$$

6.6 Unit Sample Response of Discrete-Time Systems

Example of a Non-causal System

Moving Average Filter: To observe the slow change trend over a period of time, a moving smoothing system can be used to filter out high-frequency components.

$$y(n) = \frac{1}{2M+1} \sum_{k=-M}^M x(n-k)$$



Given that the unit sample response of a system is $h(n) = a^n u(n)$, the system is ().

- ☐ A Causal and stable
- ☐ B Non-causal and stable
- ☒ C Causal; stable when $|a| < 1$, unstable when $|a| > 1$
- ☐ D Causal; unstable when $|a| < 1$, stable when $|a| > 1$

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6.6 Unit Sample Response of Discrete-Time Systems

Example 6-7: Given the unit sample response of a system $h(n) = a^n u(n)$, determine its causality and stability.

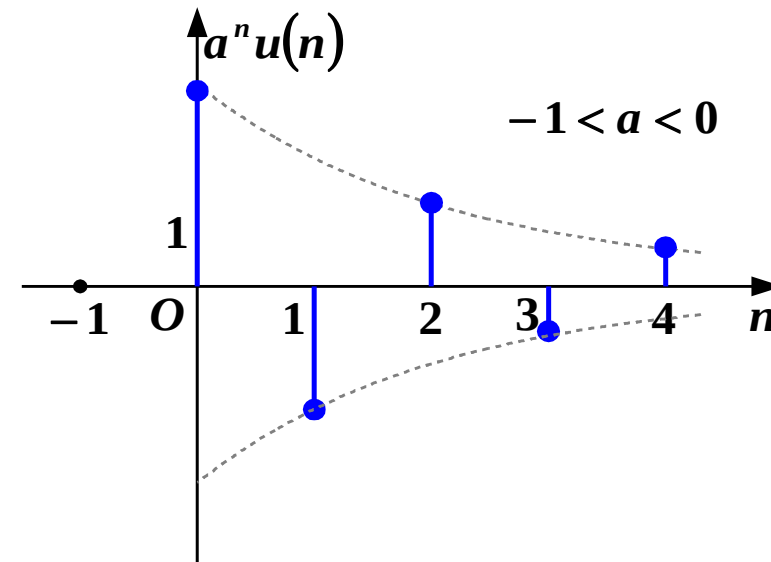
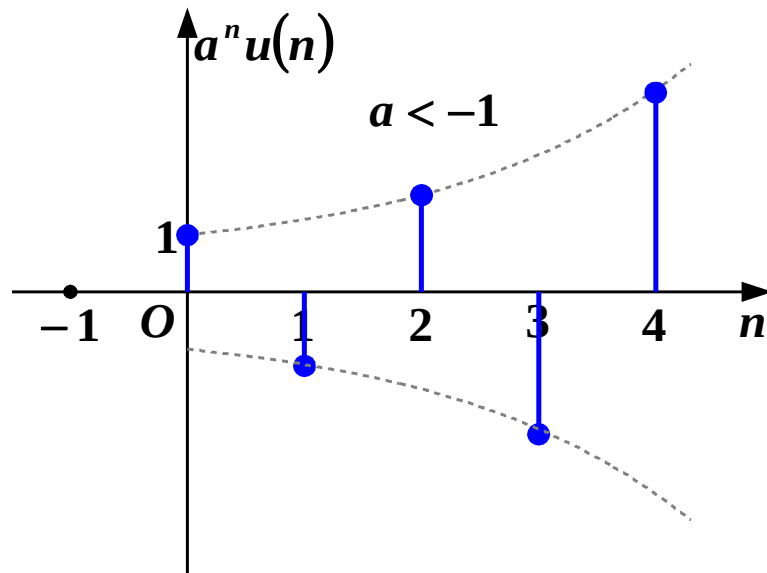
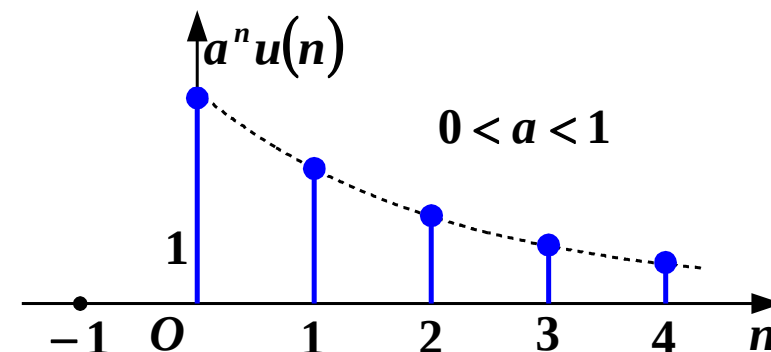
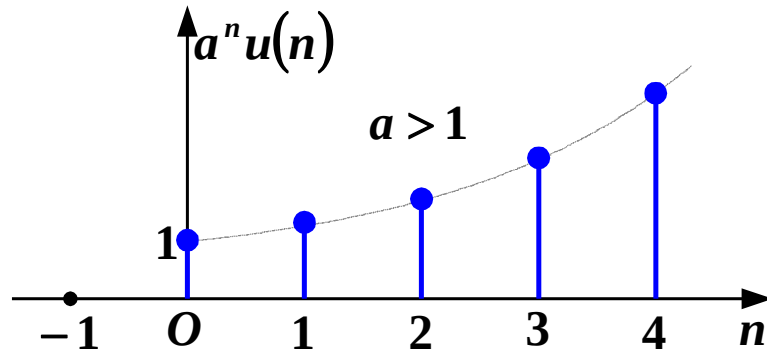
Solution: $n < 0, u(n) = 0 \quad \therefore n < 0, h(n) = a^n u(n) = 0$

Thus, the system is a **causal system**.

$$\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=0}^{\infty} |a|^n = \begin{cases} |a| < 1 & \frac{1}{1-|a|} & \text{bounded and stable} \\ |a| > 1 & \lim_{n \rightarrow \infty} \frac{1-|a|^{n+1}}{1-|a|} & \text{divergent and unstable} \end{cases}$$

6.6 Unit Sample Response of Discrete-Time Systems

When $|a| > 1$ the sequence diverges; when $|a| < 1$ the sequence converges.

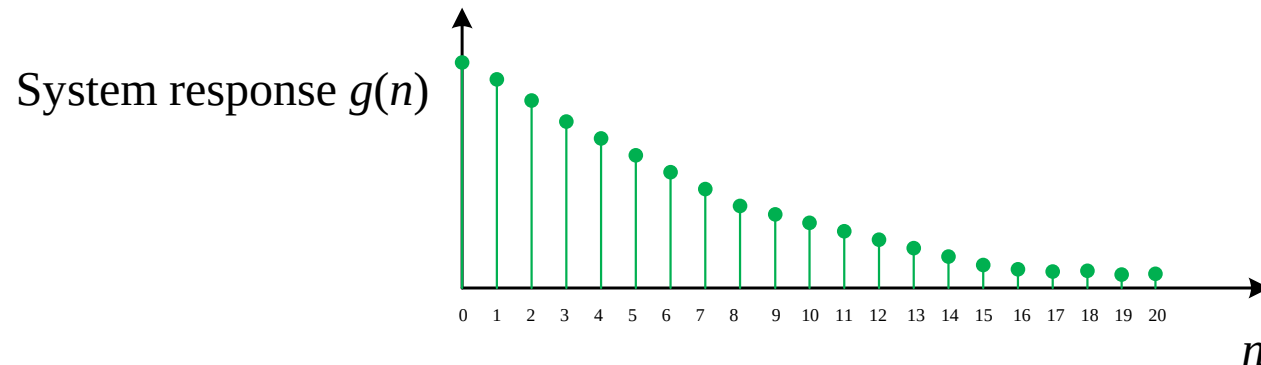
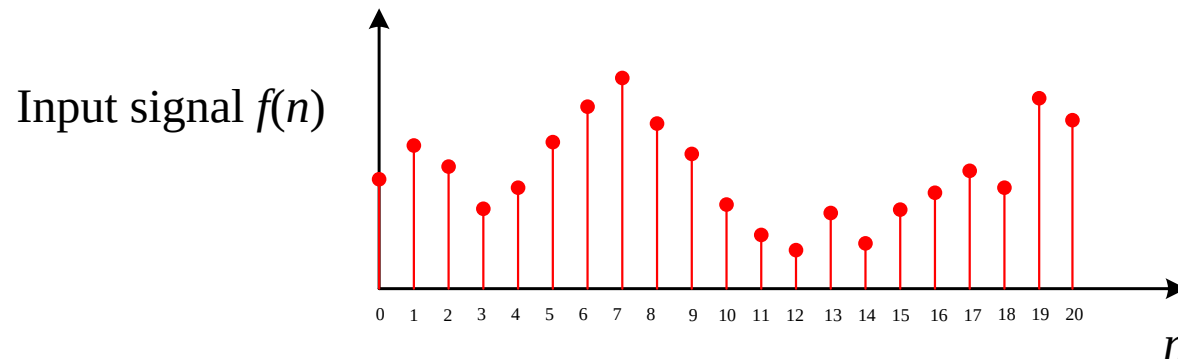


6.7 Convolution Sum

6.7.1 Definition and Physical Meaning of Convolution Sum

The convolution sum of any signals $x_1(n)$ and $x_2(n)$ is defined as :

$$x(n) = x_1(n) * x_2(n) = \sum_{m=-\infty}^{\infty} x_1(m)x_2(n-m)$$

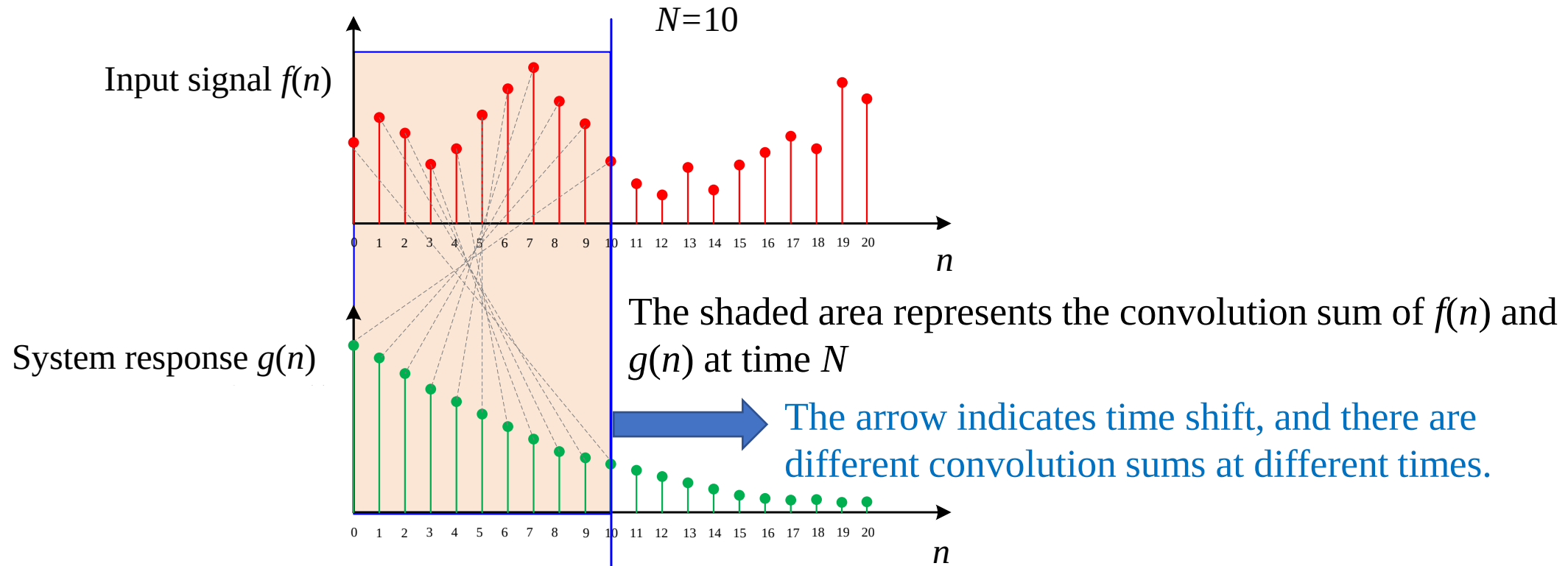


$f(n)$ —Input signal.

$g(n)$ —Represents the attenuation trend of the input signal value at a certain moment.

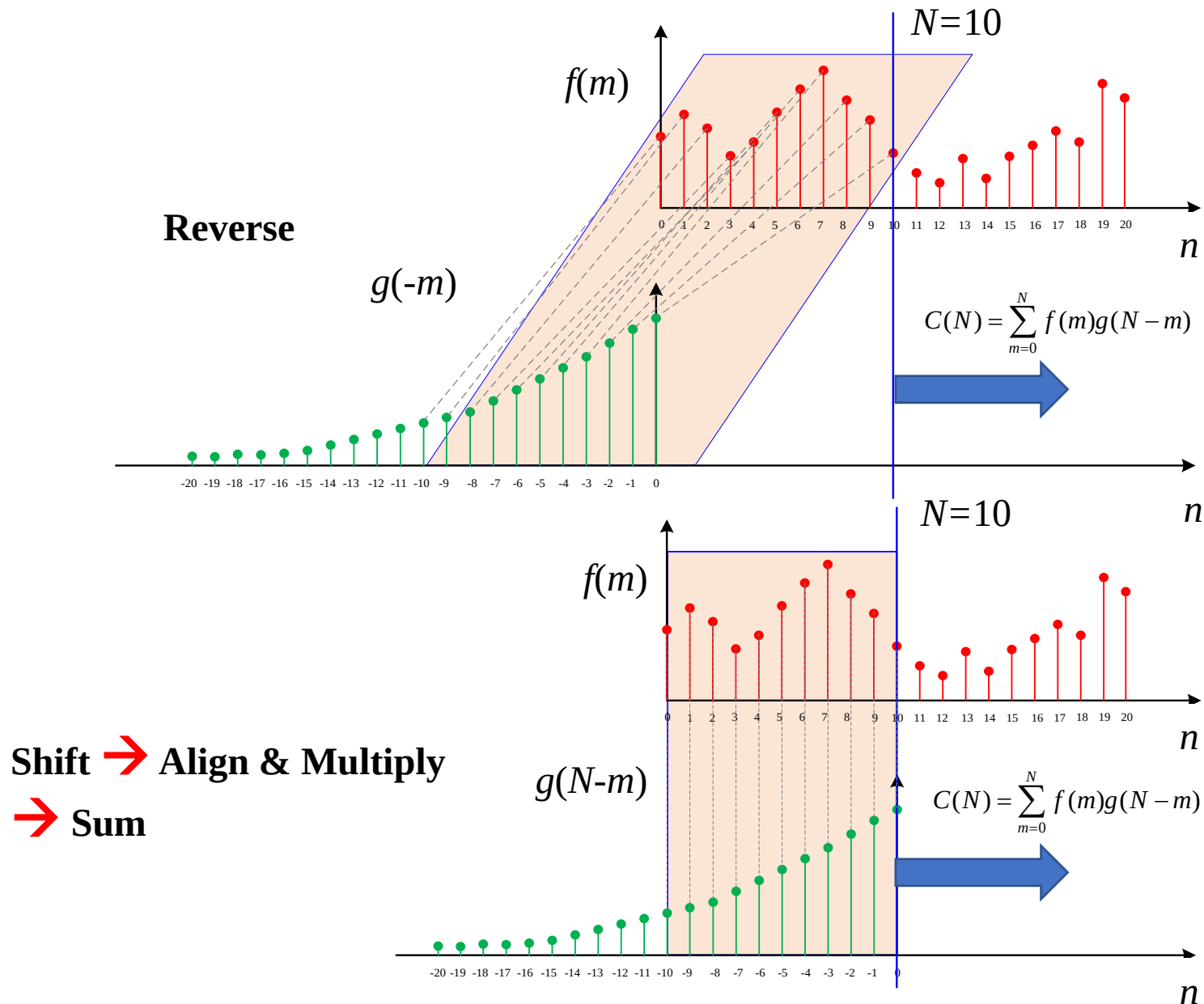
At time N , the input $f(0)$ at $n=0$ attenuates to $f(0)g(N)$.

6.7 Convolution Sum



The final output of the system reflects the **cumulative effect** of all previous input signals. At $N=10$, the output is related to the entire **shaded area**. Since $f(10)$ is just input, its contribution is $f(10)g(0)$; $f(9)$ has attenuated for 1 time unit, contributing $f(9)g(1)$ and so on. Summing these products $f(10)g(0) + f(9)g(1) + \dots + f(0)g(10)$ gives the output at $N=10$, which is the **convolution sum** of $f(n)$ and $g(n)$ at $N=10$.

6.7 Convolution Sum



Core Steps of Convolution (The “Four-Step Process”)

- 1. Reverse:** Transform $g(m)$ to $g(-m)$;
- 2. Shift:** Shift $g(-m)$ by t to get $g(n-m)$;
- 3. Align and Multiply:** Multiply the overlapping segments of $f(m)$ and $g(N-m)$;
- 4. Sum:** Compute the sum of the product over m to get the convolution value at time N .

6.7 Convolution Sum

6.7.2 Finding Zero-State Response Using Convolution Sum

Decompose the excitation signal into a linear combination of unit sample sequences. Using the properties of LTI systems, find the response of each sample sequence acting on the system individually, and superimpose these responses to obtain the **zero-state response** under any excitation.

Any excitation can be expressed as a weighted sum of unit samples:

$$x(n) = \sum_{m=-\infty}^{\infty} x(m)\delta(n-m)$$

$$\delta(n) \Rightarrow h(n) \quad \text{From time-invariance: } \delta(n-m) \Rightarrow h(n-m)$$

$$\begin{aligned} \text{From linearity: } y(n) &= T \left\{ \sum_{m=-\infty}^{\infty} x(m)\delta(n-m) \right\} = \sum_{m=-\infty}^{\infty} x(m)h(n-m) \\ &= \sum_{m=-\infty}^{\infty} h(m)x(n-m) = x(n) * h(n) \end{aligned}$$

The zero-state response of a discrete-time LTI system is equal to the convolution sum of the excitation signal and the unit sample response.

6.7 Convolution Sum

6.7.3 Operation Method of Convolution Sum

$$y(n) = x(n) * h(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m)$$

The graphical method of convolution sum consists of five steps:

(1) Variable Substitution: Change the independent variable in $x(n)$ and $h(n)$ from n to m , where m becomes the independent variable of the function.

(2) Reversing: Fold one of the signals (usually the simpler one)

$$h(m) \xrightarrow{\text{Reverse}} h(-m)$$

(3) Shifting: Shift the folded signal

$$h(-m) \xrightarrow{\text{shift by } n} h(-(m-n)) = h(n-m)$$

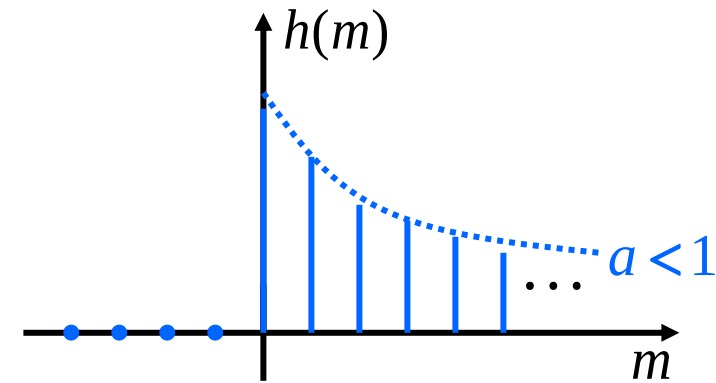
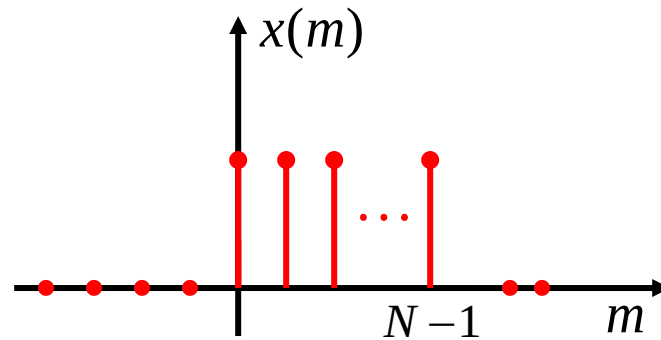
(4) Multiplying: Multiply $x(m)$ by $h(n-m)$

(5) Summing: Sum the product graph.

6.7 Convolution Sum

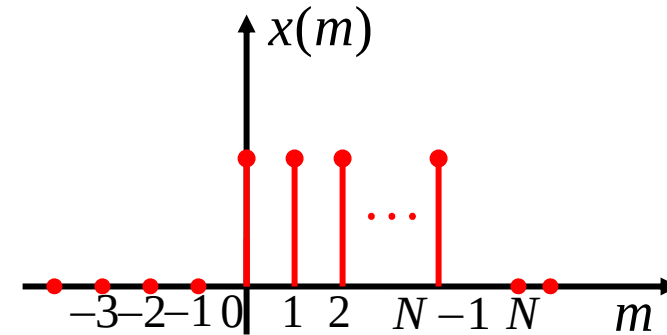
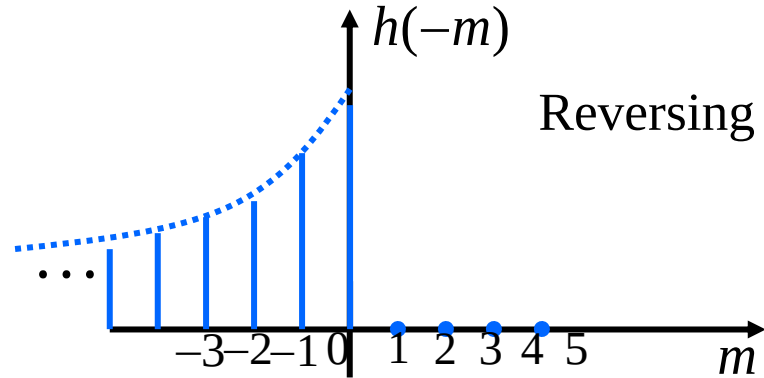
Example 6-8: Given the unit sample response of a system $h(n] = a^n u(n]$ where $0 < a < 1$, and the excitation signal $x(n] = R_N(n] = u(n] - u(n - N)$, find the zero-state response.

Solution:

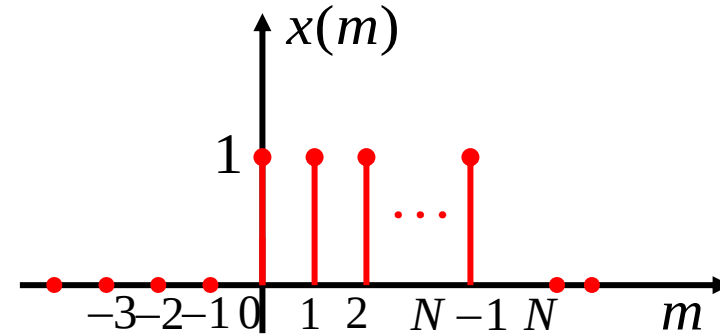
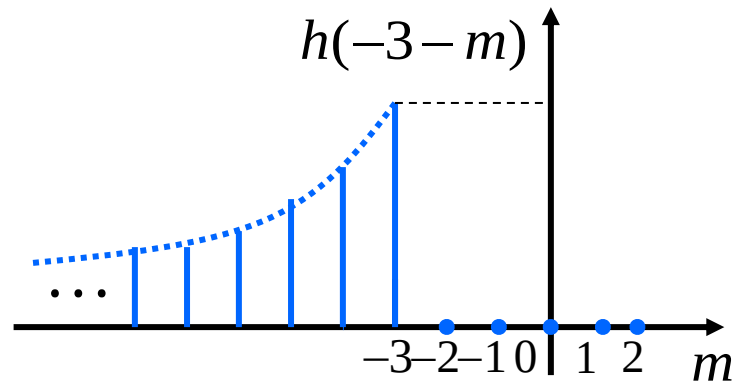


$$y_{zs}(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m) = \sum_{m=-\infty}^{\infty} [u(m) - u(m-N)]a^{n-m}u(n-m)$$

6.7 Convolution Sum



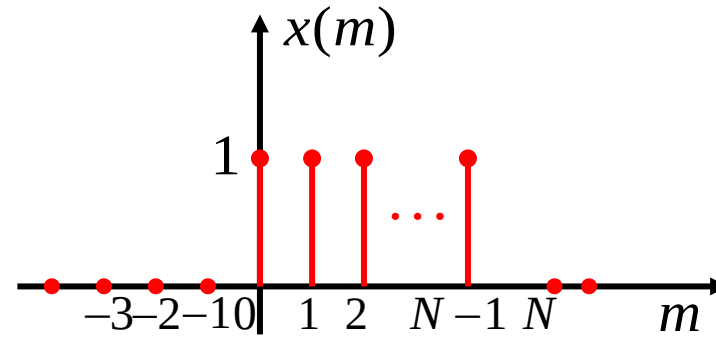
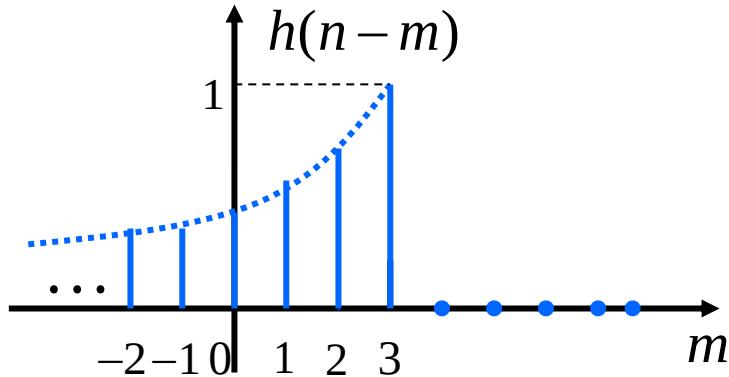
(1) when $n < 0$, $x(m)$ and $h(n-m)$ do not overlap, so $y_{zs}(n) = 0$ ($n < 0$)



6.7 Convolution Sum

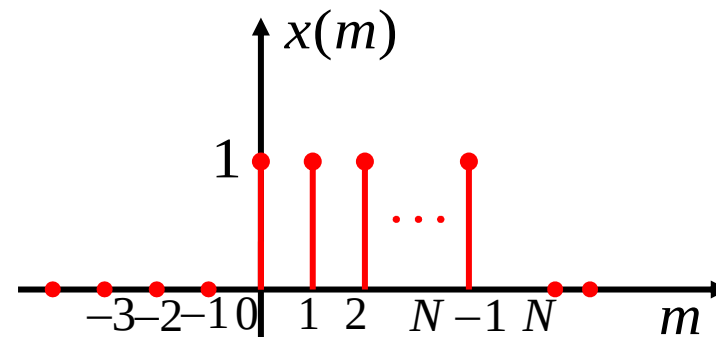
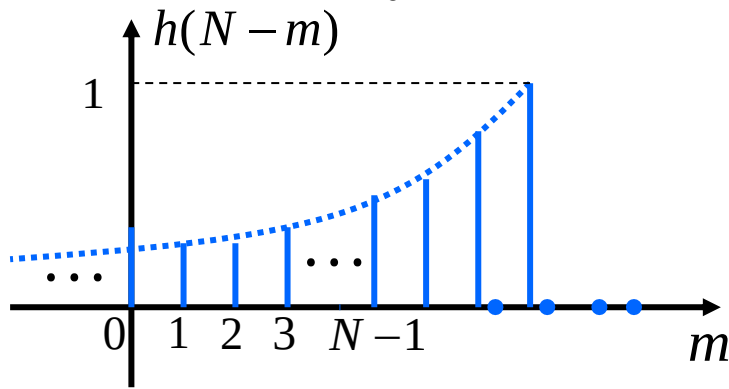
(2) When $0 \leq n < N - 1$, $x(m)$ and $h(n - m)$ overlap in the interval $0 \sim n$

$$y_{zs}(n) = \sum_{m=0}^n a^{n-m} = a^n \sum_{m=0}^n a^{-m} = a^n \frac{1 - a^{-(n+1)}}{1 - a^{-1}} \quad (0 \leq n < N - 1)$$

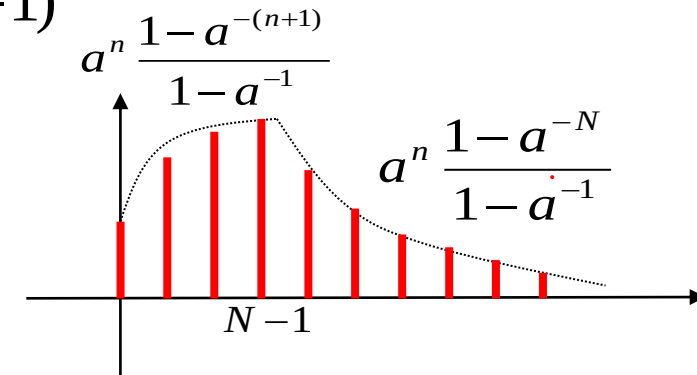


(3) When $n \geq N - 1$, $x(m)$ and $h(n - m)$ overlap in the interval $0 \sim (N - 1)$

$$y_{zs}(n) = \sum_{m=0}^{N-1} a^{n-m} = a^n \frac{1 - a^{-N}}{1 - a^{-1}} \quad (n \geq N - 1)$$



The result of convolution sum



6.7 Convolution Sum

Theorem: Given a discrete-time LTI system, its unit sample response $h(n)$ has non-zero values only in the interval $N_0 \leq n \leq N_1$, and the input $x(n)$ has non-zero values only in $N_2 \leq n \leq N_3$. Then, the zero-state response has non-zero values only in $N_4 \leq n \leq N_5$, where

$$N_4 = N_0 + N_2 \quad N_5 = N_1 + N_3$$

Proof: Let $y(n) = h(n) * x(n) = \sum_{m=-\infty}^{\infty} h(m)x(n-m)$

$$N_0 \leq m \leq N_1$$

$$N_2 \leq n - m \leq N_3$$

Add the two inequalities above.

$$N_0 + N_2 \leq n \leq N_1 + N_3$$

$$N_4 = N_0 + N_2 \quad N_5 = N_1 + N_3$$

6.7 Convolution Sum

Assume the unit sample response $\mathbf{h(n)}$ has non-zero values only in $N_0 \leq n \leq N_1$, with sequence length $L_1 = N_1 - N_0 + 1$. The input $\mathbf{x(n)}$ has non-zero values only in $N_2 \leq n \leq N_3$, with sequence length $L_2 = N_3 - N_2 + 1$.

The zero-state response $\mathbf{y(n)}$ has non-zero values only in $N_0 + N_2 \leq n \leq N_1 + N_3$, with **sequence length L_3** :

$$\begin{aligned} L_3 &= (N_1 + N_3) - (N_0 + N_2) + 1 \\ &= (N_1 - N_0) + (N_3 - N_2) + 1 \\ &= (N_1 - N_0 + 1) + (N_3 - N_2 + 1) - 1 \end{aligned}$$

$$L_3 = L_1 + L_2 - 1$$

For continuous-time signals, the length of the convolution integral is $L_3 = L_1 + L_2$.

6.7 Convolution Sum

Example 6-9: Given $x_1(n) = 2\delta(n) + \delta(n-1) + 4\delta(n-2) + \delta(n-3)$

$$x_2(n) = 3\delta(n) + \delta(n-1) + 5\delta(n-2)$$

Find the convolution $y(n) = x_1(n) * x_2(n)$.

Solution: $\{x_1(n)\} = \{ \overset{\downarrow}{2} \quad 1 \quad 4 \quad 1 \}$ $\{x_2(n)\} = \{ \overset{\downarrow}{3} \quad 1 \quad 5 \}$

$$x_1(n) : 2 \quad 1 \quad 4 \quad 1$$

$$x_2(n) : \quad 3 \quad 1 \quad 5$$

$$10 \quad 5 \quad 20 \quad 5$$

$$2 \quad 1 \quad 4 \quad 1$$

$$6 \quad 3 \quad 12 \quad 3$$

$$y(n) : 6 \quad 5 \quad 23 \quad 12 \quad 21 \quad 5$$

**Perform the term-by-term
multiplication and summation**

**no carryover during
multiplication and summation**

$$\{y(n)\} = \{ \overset{\downarrow}{6} \quad 5 \quad 23 \quad 12 \quad 21 \quad 5 \}$$

Calculate the convolution sum of $x(n) = \{1, 2, \overset{\downarrow}{0}, 3, 2\}$ and $h(n) = \{1, \overset{\downarrow}{4}, 2, 3\}$

A $y(n) = \{1, 6, \overset{\downarrow}{10}, 10, 20, 14, 13, 6\}$

B $y(n) = \{1, 8, \overset{\downarrow}{10}, 10, 21, 14, 13, 6\}$

☒ C $y(n) = \{1, 6, 10, \overset{\downarrow}{10}, 20, 14, 13, 6\}$

D $y(n) = \{1, 8, 10, \overset{\downarrow}{10}, 21, 14, 13, 6\}$

提交

6.7 Convolution Sum

6.7.4 Properties of Convolution Sum

Commutative Law

$$x_1(n) * x_2(n) = x_2(n) * x_1(n)$$

Distributive Law

$$x_1(n) * [x_2(n) + x_3(n)] = x_1(n) * x_2(n) + x_1(n) * x_3(n)$$

Associative Law

$$x_1(n) * [x_2(n) * x_3(n)] = [x_1(n) * x_2(n)] * x_3(n)$$

Convolution with Unit Sample Sequence

$$x(n) * \delta(n) = x(n)$$

$$x(n) * \delta(n - n_0) = x(n - n_0)$$

Convolution with Step Sequence

$$x(n) * u(n) = \sum_{m=-\infty}^n x(m)$$

6.7 Convolution Sum

6.7.5 Finding Zero-State Response of a System Using Convolution Sum

Example 6-10: The difference equation of a discrete-time LTI system is

$$y(n) + 3y(n-1) + 2y(n-2) = x(n)$$

Given the excitation $x(n) = 3\left(\frac{1}{2}\right)^n u(n)$, find the zero-state response $y_{zs}(n)$.

Solution:

Step 1: Find the unit sample response of the system

$$h(n) + 3h(n-1) + 2h(n-2) = \delta(n)$$

The characteristic roots are **-1 & -2**, $\therefore h(n) = \left[C_1 (-1)^n + C_2 (-2)^n \right] u(n)$

$h(-1) = 0$, $h(-2) = 0$, iterate to find $h(0) = 1$.

Using $h(0)$ and $h(-1)$, get $C_1 = -1$, $C_2 = 2$.

$$\therefore h(n) = \left[-(-1)^n + 2(-2)^n \right] u(n)$$

6.7 Convolution Sum

Step 2: Find the convolution sum of the excitation signal and the unit sample response to obtain the zero-state response.

$$\begin{aligned}y_{zs}(n) &= \sum_{m=-\infty}^{\infty} x(m)h(n-m) \\&= \sum_{m=-\infty}^{\infty} 3\left(\frac{1}{2}\right)^m u(m) \cdot [-(-1)^{n-m} + 2(-2)^{n-m}] u(n-m) \\&= \begin{cases} -3(-1)^n \sum_{m=0}^n \left(-\frac{1}{2}\right)^m + 6(-2)^n \sum_{m=0}^n \left(-\frac{1}{4}\right)^m, & n \geq 0 \\ 0 & n < 0 \end{cases} \\&= [-2(-1)^n + \frac{24}{5}(-2)^n + \frac{1}{5}\left(\frac{1}{2}\right)^n] u(n)\end{aligned}$$

1. Repeat all the examples in the lecture notes.
2. 2.32(a), 2.34(b), 2.49(f)(g)(i)(j) (judge only causality and stability in 2.49) in the textbook.